

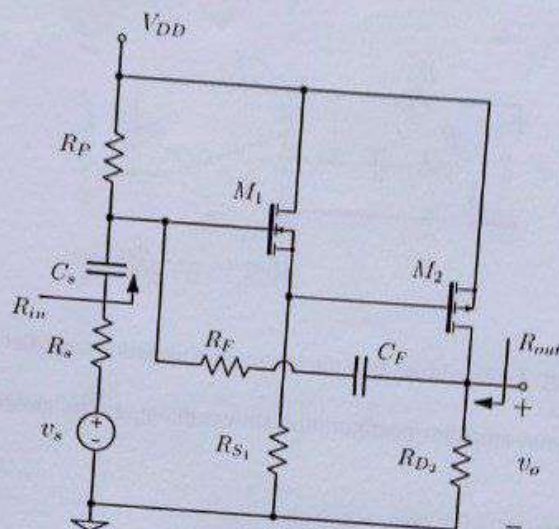


Fig. 1 – A case of two-stage CMOS amplifier.

Consider the circuit in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{DD} = 5\text{ V}$; $R_P = 100\text{ k}\Omega$; $R_S = 10\text{ k}\Omega$; $R_{S1} = 2.2\text{ k}\Omega$; $R_{D2} = 1.5\text{ k}\Omega$; $R_F = 100\text{ k}\Omega$; $C_F = 1\text{ }\mu\text{F}$; $C_S = 100\text{ nF}$.

M_1 (e-NMOS): $V_{t1} = +0.5\text{ V}$; $I_{DSS} = 0.6\text{ mA}$; $\lambda_n = 0$;

M_2 (e-PMOS): $V_{t2} = -0.5\text{ V}$; $I_{SDS} = 0.6\text{ mA}$; $\lambda_p = 0$.

Considering all capacitors to be equivalent to open circuits, determine:

1. the operating points (V_{DS} , I_D) of M_1 and M_2 ;
2. the power drawn by the amplifier from the supply.

Assuming that, at the frequencies of interest (i.e. at mid-band), C_S and C_F are equivalent to short circuits, determine:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance, R_{in} , indicated in the figure;
5. the output resistance, R_{out} , indicated in the figure.

R_{B1}

R_{B2}

$$R_{th} = 15 + k\Omega = 15$$

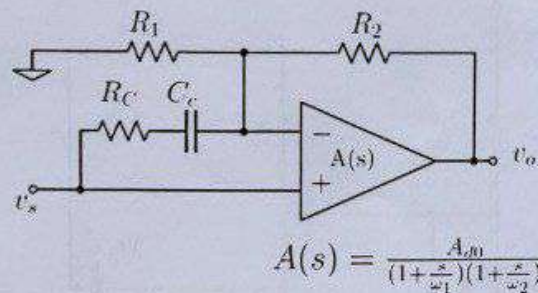


Fig. 2 – Non ideal, OpAmp based, non inverting amplifier.

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 33 \text{ k}\Omega$; $R_2 = 120 \text{ k}\Omega$; $R_C = 150 \text{ }\Omega$; $C_C = 330 \text{ nF}$;

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$; $\omega_1 = 2 \text{ krad/s}$; $\omega_2 = 600 \text{ krad/s}$.

Considering, initially, C_C to be open circuited, determine:

- ☒ the expression of the *ideal* amplifier voltage gain $A_v(s) = v_o/v_s$;
- ☒ the equivalent *block diagram* of the amplifier, according to *feedback theory*;
- ☒ the amplifier *bandwidth* and *phase margin*.

Then, applying the noise gain compensation network R_C - C_C , find:

4. the *bandwidth* and the *phase margin* of the compensated amplifier.